# Port Scan Activity [Let's Defend – Investigate]

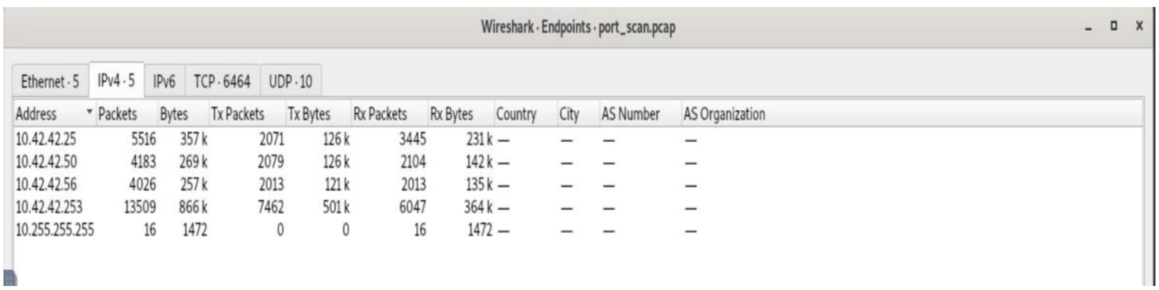
Can you determine evidences of port scan activity?

Log file: /root/Desktop/ChallengeFile/port_scan.pcap

Note: pcap file found public resources.

## | Start Investigation

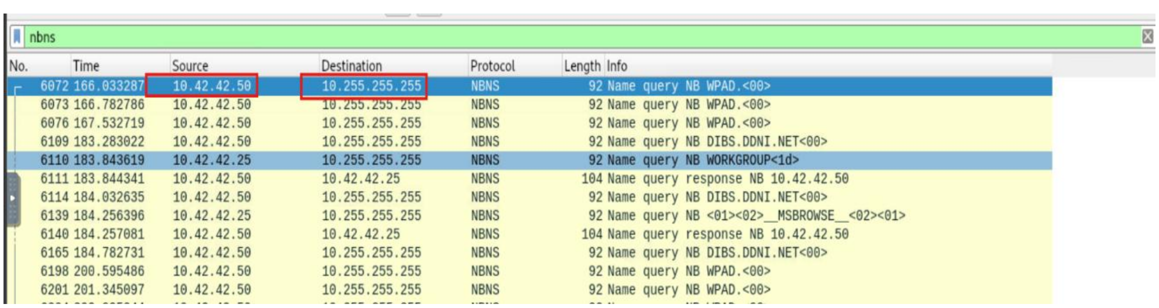
### Q1: What is the IP address scanning the environment?




The scanning IP address is 10.42.42.253. I identified it because it has the highest number of transmitted packets (13,509 packets) and the largest amount of data traffic compared to other hosts in the network.

Answer: 10.42.42.253

### Q2: What is the IP address found as a result of the scan?




The scanning process revealed the IP address 10.42.42.50. This was confirmed by observing multiple NBNS name queries sent from this host to the broadcast address (10.255.255.255) to discover other devices on the network.

Answer: 10.42.42.50

## Q3: What is the MAC address of the Apple system it finds?

Head to **Statistics** –> **Conversations** –> **Ethernet** tab and toggle the **Name Resolution** option at the bottom. We are able to find a Apple system.

Wireshark · Conversations · port_scan.pcap

Ethernet · 6 | IPv4 · 6 | IPv6 | TCP · 7436 | UDP · 12

| Address A | Address B | Packets | Bytes | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Apple_92:6e:dc | QuantaCo_82:1f:4a | 5412 | 348 k | 2007 | 120 k | 3405 | 228 k | 0.607596 | 603.2091 | 1600 | 3024 |
| Apple_92:6e:dc | Broadcast | 4 | 368 | 4 | 368 | 0 | 0 | 183.843619 | 360.4177 | 8 | 0 |
| Apple_92:6e:dc | CompalIn_51:d7:b2 | 100 | 7964 | 60 | 4968 | 40 | 2996 | 183.844341 | 360.7820 | 110 | 66 |
| QuantaCo_82:1f:4a | CompalIn_51:d7:b2 | 4071 | 259 k | 2044 | 137 k | 2027 | 122 k | 0.000000 | 603.9423 | 1823 | 1619 |
| QuantaCo_82:1f:4a | CompalIn_cb:1e:79 | 4026 | 257 k | 2013 | 135 k | 2013 | 121 k | 0.607594 | 605.4745 | 1795 | 1610 |
| CompalIn_51:d7:b2 | Broadcast | 12 | 1104 | 12 | 1104 | 0 | 0 | 166.033287 | 53.3117 | 165 | 0 |

Wireshark · Conversations · port_scan.pcap

Ethernet · 6 | IPv4 · 6 | IPv6 | TCP · 7436 | UDP · 12

| Address A | Address B | Packets | Bytes | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 00:16:cb:92:6e:dc | 00:23:8b:82:1f:4a | 5412 | 348 k | 2007 | 120 k | 3405 | 228 k | 0.607596 | 603.2091 | 1600 | 3024 |
| 00:16:cb:92:6e:dc | ff:ff:ff:ff:ff:ff | 4 | 368 | 4 | 368 | 0 | 0 | 183.843619 | 360.4177 | 8 | 0 |
| 00:16:cb:92:6e:dc | 70:5a:b6:51:d7:b2 | 100 | 7964 | 60 | 4968 | 40 | 2996 | 183.844341 | 360.7820 | 110 | 66 |
| 00:23:8b:82:1f:4a | 70:5a:b6:51:d7:b2 | 4071 | 259 k | 2044 | 137 k | 2027 | 122 k | 0.000000 | 603.9423 | 1823 | 1619 |
| 00:23:8b:82:1f:4a | 00:26:22:cb:1e:79 | 4026 | 257 k | 2013 | 135 k | 2013 | 121 k | 0.607594 | 605.4745 | 1795 | 1610 |
| 70:5a:b6:51:d7:b2 | ff:ff:ff:ff:ff:ff | 12 | 1104 | 12 | 1104 | 0 | 0 | 166.033287 | 53.3117 | 165 | 0 |

Answer: 00:16:cb:92:6e:dc

## Q4: What is the IP address of the detected Windows system?

Head to *Statistics –> Endpoint –> IPv4*

From what we gather so far:

- 10.42.42.25 – Apple System
- 10.42.42.50 – Windows System
- 10.42.42.56 – Unknown System
- 10.42.42.253 – Attacking System
- 10.255.255.255 – Network Broadcast Address

Wireshark · Conversations · port_scan.pcap

Ethernet · 6 | IPv4 · 6 | IPv6 | TCP · 7436 | UDP · 12

| Address A | Address B | Packets | Bytes | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 10.42.42.25 | 10.42.42.253 | 5412 | 348 k | 2007 | 120 k | 3405 | 228 k | 0.607596 | 603.2091 | 1600 | 3024 |
| 10.42.42.25 | 10.255.255.255 | 4 | 368 | 4 | 368 | 0 | 0 | 183.843619 | 360.4177 | 8 | 0 |
| 10.42.42.25 | 10.42.42.50 | 100 | 7964 | 60 | 4968 | 40 | 2996 | 183.844341 | 360.7820 | 110 | 66 |
| 10.42.42.50 | 10.42.42.253 | 4071 | 259 k | 2027 | 122 k | 2044 | 137 k | 0.000000 | 603.9423 | 1619 | 1823 |
| 10.42.42.50 | 10.255.255.255 | 12 | 1104 | 12 | 1104 | 0 | 0 | 166.033287 | 53.3117 | 165 | 0 |
| 10.42.42.56 | 10.42.42.253 | 4026 | 257 k | 2013 | 121 k | 2013 | 135 k | 0.607594 | 605.4745 | 1610 | 1795 |

Answer: 10.42.42.50